# CNG KSP Feature Matrix

Every capability observed across shipping CNG Key Storage Providers, with this project's coverage and — where a gap exists — what closing it would take. Generated 2026-09-09.

| 96 | 44 | 5 | 47 | 46% | 44 |
|---|---|---|---|---|---|
| Capabilities tracked | Covered | Partial | Not covered | Fully covered | Actionable gaps |

## How to read this

| Status | Meaning |
|---|---|
| Covered | Implemented and exercised by the test suite. |
| Partial | Implemented, but reachable only through a non-standard identifier or with a stated caveat. |
| Not covered | Absent. The Gap column says what closing it would take. |

| Ev | Evidence for the market claim |
|---|---|
| A | Read from source code or an authoritative API reference that enumerates the provider's own algorithm list. |
| B | Stated in vendor or Microsoft documentation surfaced through search; primary page not read in full. |
| C | Drawn from HSM datasheets describing the device rather than its KSP. Treat as a lead, not a fact. |

**Effort** on gaps: S = a few lines to a day · M = days · L = a substantial piece of work. **N/A** marks a deliberate position or an external blocker rather than a backlog item — those are excluded from the actionable-gap count.

| ID | Feature | Detail | Market support | Ev | Status | Gap — what closing it would take | Eff |
|---|---|---|---|---|---|---|---|
| **PROVIDER INTERFACE** | | | | | | | |
| IFACE-01 | **GetKeyStorageInterface export** | Sole exported symbol; returns the KSP function table | All providers | A | Covered | — | — |
| IFACE-02 | **OpenProvider / FreeProvider** | Provider handle lifecycle | All providers | A | Covered | — | — |
| IFACE-03 | **GetProviderProperty** | Name / Version / ImplType | All providers | A | Covered | — | — |
| IFACE-04 | **SetProviderProperty** | Provider-level configuration | Vendor-specific | C | Not covered | Stubbed as NTE_NOT_SUPPORTED. Would need a defined config surface — slot selection and PIN source. Low value until multi-slot exists. | M |
| IFACE-05 | **NotifyChangeKey** | Key-change notification handle | Microsoft KSP | B | Not covered | Returns ERROR_SUCCESS as a no-op. Real support needs a registry/file watcher on the token store and an event object per caller. | L |
| IFACE-06 | **GetOperationProperty** | Async / overlapped operation state | Microsoft KSP | B | Not covered | Returns NTE_NOT_SUPPORTED. Only needed for overlapped I/O; PKCS#11 is synchronous so there is nothing to report. | L |
| IFACE-07 | **PromptUser** | Interactive PIN / consent dialog | Smart card KSPs | B | Not covered | Returns NTE_NOT_SUPPORTED. Would need a UI thread and a window handle from NCRYPT_■WINDOW_■HANDLE_■PROPERTY. | L |
| IFACE-08 | **FreeBuffer / FreeObject** | Provider-allocated memory release | All providers | A | Covered | — | — |
| **ASYMMETRIC RSA** | | | | | | | |
| RSA-01 | **RSA key generation** | 2048 / 3072 / 4096 bits | All providers | A | Covered | — | — |
| RSA-02 | **RSA 1024 and below** | 512-1024 bits | MS Software KSP; Utimaco | B | Not covered | Deliberately rejected with NTE_BAD_LEN. Would be a one-line change in ksp_properties.c but weakens the security posture; leave as is unless a legacy CA demands it. | S |

| ID | Feature | Detail | Market support | Ev | Status | Gap — what closing it would take | Eff |
|---|---|---|---|---|---|---|---|
| RSA-03 | **RSA above 4096** | 8192 / 16384 bits | MS Software KSP; Utimaco | B | Not covered | Add sizes to the validation list in ksp_properties.c. SoftHSM2 supports them but generation is very slow (minutes for 16384). | S |
| RSA-04 | **RSA arbitrary size increments** | 64-bit (MS) or 8-bit (Utimaco) steps | MS Software KSP; Utimaco | B | Not covered | Replace the fixed {2048 3072 4096} set with a range check. Trivial; the discrete set is a deliberate choice. | S |
| RSA-05 | **Custom public exponent** | Exponent other than 65537 | Vendor-specific | C | Not covered | Hardcoded to 65537 in ksp_key.c. Would read NCRYPT_■PUBLIC_■EXPONENT if any caller needed it; no standard CNG property exists for this. | S |
| RSA-06 | **PKCS#1 v1.5 signing** | SHA-1 / 256 / 384 / 512 | All providers | A | Covered | — | — |
| RSA-07 | **PSS signing** | SHA-1 / 256 / 384 / 512 with caller salt length | All providers | B | Covered | — | — |
| RSA-08 | **PSS with SHA-224** | SHA-224 salt and MGF1 | Unknown | C | Not covered | P11_MapHashAlg already maps SHA-224; only FillPssParams omits it. One branch in ksp_crypto.c. | S |
| RSA-09 | **PKCS#1 v1.5 decryption** | AT_KEYEXCHANGE keys | All providers | A | Covered | — | — |
| RSA-10 | **OAEP decryption** | SHA-1 / 224 / 256 / 384 / 512 | All providers | A | Covered | — | — |
| RSA-11 | **OAEP application label** | pbLabel / cbLabel passed through | Unknown | C | Covered | — | — |
| RSA-12 | **Raw RSA (no padding)** | CKM_RSA_X_509 | Some HSMs | C | Not covered | SoftHSM2 does not implement CKM_RSA_X_509. Blocked by the backend not by the KSP; would work against an HSM that offers it. | M |
| **ASYMMETRIC ECDSA** | | | | | | | |
| ECDSA-01 | **ECDSA P-256** | secp256r1; 64-byte signature | All providers | A | Covered | — | — |
| ECDSA-02 | **ECDSA P-384** | secp384r1; 96-byte signature | All providers | A | Covered | — | — |
| ECDSA-03 | **ECDSA P-521** | secp521r1; 132-byte signature | All providers | A | Covered | — | — |
| ECDSA-04 | **Brainpool curves** | brainpoolP256r1 / 384r1 / 512r1 | Entrust; Utimaco (device) | C | Not covered | Add the DER OIDs to config.h and P11_GetCurveOid. SoftHSM2 supports them. CNG has BCRYPT_■ECC_■CURVE_■BRAINPOOL* names so this stays standards-compliant. | M |
| ECDSA-05 | **secp256k1** | Bitcoin curve | Entrust (device) | C | Not covered | Same mechanism as ECDSA-04: one OID entry. CNG names it BCRYPT_■ECC_■CURVE_■SECP256K1. | S |
| ECDSA-06 | **DER to r\|\|s conversion** | PKCS#11 returns DER; CNG wants raw | All providers | A | Covered | — | — |
| **KEY AGREEMENT** | | | | | | | |
| ECDH-01 | **ECDH P-256 / P-384 / P-521** | CKM_ECDH1_DERIVE with CKD_NULL | MS Software KSP; Utimaco | B | Covered | — | — |
| ECDH-02 | **SecretAgreement / DeriveKey / FreeSecret** | NCRYPT_■SECRET_■HANDLE lifecycle | MS Software KSP; Utimaco | B | Covered | — | — |
| ECDH-03 | **BCRYPT_KDF_RAW_SECRET** | Raw Z value returned | MS Software KSP | B | Covered | — | — |
| ECDH-04 | **BCRYPT_KDF_HASH** | SP 800-56A concatenation KDF | MS Software KSP | B | Not covered | Returns NTE_NOT_SUPPORTED. Implement by hashing the Z value with the caller-supplied algorithm and info from the NCryptBufferDesc. Moderate work; callers can run it with BCrypt instead. | M |
| ECDH-05 | **BCRYPT_KDF_HMAC / TLS_PRF / HKDF** | Protocol-specific KDFs | MS Software KSP | B | Not covered | Same shape as ECDH-04. HKDF is the one worth doing first for TLS 1.3. | M |
| ECDH-06 | **Finite-field Diffie-Hellman** | DH 512-4096 | MS Software KSP | B | Not covered | Out of scope by choice. ECDH supersedes it and no surveyed HSM KSP exposes it. Would need CKM_■DH_■PKCS_■KEY_■PAIR_■GEN plus CKM_■DH_■PKCS_■DERIVE. | L |
| **ASYMMETRIC EDDSA** | | | | | | | |
| EDDSA-01 | **Ed25519 signing** | 64-byte raw signature | None | A | Partial | Works against SoftHSM2 but uses the non-standard identifier EDDSA_ED25519. CNG defines no EdDSA algorithm ID so no stock application can request it. Nothing to fix in the KSP; blocked on Microsoft. | N/A |

| ID | Feature | Detail | Market support | Ev | Status | Gap — what closing it would take | Eff |
|---|---|---|---|---|---|---|---|
| EDDSA-02 | **Ed448 signing** | 114-byte raw signature | None | A | Partial | Same constraint as EDDSA-01. | N/A |
| EDDSA-03 | **X25519 key agreement** | Curve25519 ECDH | MS CNG has BCRYPT_ECC_CURVE_25519 | B | Not covered | This one IS standard in CNG (Windows 10+) unlike Ed25519 signing. Add the curve OID and route it through the existing ECDH path. Highest-value gap in this table. | M |
| **ASYMMETRIC DSA** | | | | | | | |
| DSA-01 | **DSA signing** | 512-1024 bits | MS Software KSP | B | Not covered | Deliberately out of scope. Deprecated in modern Windows PKI and absent from every HSM KSP surveyed. | L |
| **POST-QUANTUM** | | | | | | | |
| PQC-01 | **ML-DSA (Dilithium)** | FIPS 204 | Thales Luna (device claim) | C | Not covered | SoftHSM2 2.7.0 has no PQC mechanisms so this is blocked at the backend. CNG gained PQC identifiers in recent Windows builds; revisit when both sides support it. | L |
| PQC-02 | **ML-KEM (Kyber)** | FIPS 203 | Thales Luna (device claim) | C | Not covered | Same blocker as PQC-01. | L |
| PQC-03 | **LMS / XMSS** | Stateful hash-based signatures | Thales Luna (device claim) | C | Not covered | Same blocker as PQC-01. Stateful schemes also need key-state persistence the KSP has no model for. | L |
| **SYMMETRIC AES** | | | | | | | |
| AES-01 | **AES key generation** | 128 / 192 / 256 bits | None via KSP | A | Covered | — | — |
| AES-02 | **AES-ECB** | No IV | None via KSP | A | Covered | — | — |
| AES-03 | **AES-CBC** | 16-byte IV | None via KSP | A | Covered | — | — |
| AES-04 | **AES-CBC with padding** | NCRYPT_■PAD_■CIPHER_■FLAG | None via KSP | A | Covered | — | — |
| AES-05 | **AES-GCM** | 12-byte nonce; 128-bit tag; AAD | None via KSP | A | Covered | — | — |
| AES-06 | **AES-CTR** | 16-byte counter block | None via KSP | A | Partial | Works but ChainingModeCTR is a KSP extension. CNG defines no standard counter-mode string so no stock application will select it. | N/A |
| AES-07 | **AES-CCM** | Authenticated mode | None via KSP | A | Not covered | Returns NTE_NOT_SUPPORTED. SoftHSM2 2.7.0 does not implement CKM_AES_CCM so this is a backend blocker not a KSP one. | M |
| AES-08 | **AES-CFB** | Cipher feedback | None via KSP | A | Not covered | Returns NTE_NOT_SUPPORTED. No SoftHSM2 mechanism; would need CKM_AES_CFB128 which 2.7.0 lacks. | M |
| AES-09 | **AES key wrap** | RFC 3394 / 5649 | Some HSMs (device) | C | Not covered | SoftHSM2 offers CKM_AES_KEY_WRAP and KEY_WRAP_PAD. Needs KSP_ExportKey / ImportKey wired to C_WrapKey / C_UnwrapKey plus a CNG blob convention. | L |
| AES-10 | **AES-CMAC** | Block-cipher MAC | Some HSMs (device) | C | Not covered | SoftHSM2 offers CKM_AES_CMAC. Would route through KSP_SignHash like HMAC does; same non-standard-identifier caveat applies. | M |
| **SYMMETRIC MAC** | | | | | | | |
| HMAC-01 | **HMAC-SHA1 / 256 / 384 / 512** | Generic secret keys | None via KSP | A | Partial | Works against SoftHSM2 but HMAC_SHA* are our own identifiers. CNG does HMAC through BCrypt algorithm handles not KSP key algorithms. | N/A |
| HMAC-02 | **HMAC-SHA224** | SHA-224 variant | None | C | Not covered | SoftHSM2 has CKM_SHA224_HMAC. One entry in config.h and P11_■ResolveMechanism; inherits the same identifier caveat. | S |
| **SYMMETRIC LEGACY** | | | | | | | |
| 3DES-01 | **3DES / DES** | 56 / 112 / 168 bits | None via KSP | B | Not covered | Deliberately out of scope. Deprecated and excluded from modern HLK requirements despite SoftHSM2 supporting 14 DES mechanisms. | L |
| **KEY FORMATS** | | | | | | | |
| FMT-01 | **BCRYPT_RSAPUBLIC_BLOB export** | Header + exponent + modulus | All providers | A | Covered | — | — |

| ID | Feature | Detail | Market support | Ev | Status | Gap — what closing it would take | Eff |
|---|---|---|---|---|---|---|---|
| FMT-02 | **BCRYPT_ECCPUBLIC_BLOB export** | Header + X + Y | All providers | A | Covered | — | — |
| FMT-03 | **EdDSA public blob export** | Generic ECC magic + raw point | None | A | Covered | — | — |
| FMT-04 | **BCRYPT_RSAPUBLIC_BLOB import** | Creates a session public key object | Vendor-specific | C | Covered | — | — |
| FMT-05 | **BCRYPT_ECCPUBLIC_BLOB import** | Creates a session public key object | Vendor-specific | C | Covered | — | — |
| FMT-06 | **Private key export** | RSAFULLPRIVATE / ECCPRIVATE | None (all refuse) | A | Not covered | Refused by design with NTE_NOT_SUPPORTED. Keys are CKA_■EXTRACTABLE=FALSE. This is correct HSM behaviour not a gap. | N/A |
| FMT-07 | **Private key import** | Wrapping an external private key onto the token | Some HSMs | C | Not covered | Returns NTE_NOT_SUPPORTED. Would need C_CreateObject with private material or C_UnwrapKey. Contradicts the HSM posture; only add if a migration workflow demands it. | L |
| FMT-08 | **BCRYPT_KEY_DATA_BLOB** | Raw symmetric key material | MS Software KSP | B | Not covered | AES keys are created CKA_■EXTRACTABLE=FALSE so export cannot work. Import would need C_CreateObject with CKA_VALUE. Constants are already in the mock header. | M |
| FMT-09 | **PKCS#8 blob** | NCRYPT_■PKCS8_■PRIVATE_■KEY_■BLOB | MS Software KSP | B | Not covered | Same constraint as FMT-07: it is a private-key format and keys are non-extractable. | L |
| FMT-10 | **OPAQUETRANSPORT blob** | Vendor-opaque key transport | Vendor-specific | C | Not covered | Would let keys move between instances of the same provider. Needs a defined wrapping format; no standard to follow. | L |
| **KEY PROPERTIES** | | | | | | | |
| PROP-01 | **NCRYPT_ALGORITHM_PROPERTY** | Read | All providers | A | Covered | — | — |
| PROP-02 | **NCRYPT_LENGTH_PROPERTY** | Read and write before finalize | All providers | A | Covered | — | — |
| PROP-03 | **NCRYPT_KEY_TYPE_PROPERTY** | AT_SIGNATURE / AT_KEYEXCHANGE | All providers | A | Covered | — | — |
| PROP-04 | **NCRYPT_NAME / UNIQUE_NAME** | Key label | All providers | A | Covered | — | — |
| PROP-05 | **NCRYPT_EXPORT_POLICY_PROPERTY** | Always 0 (non-exportable) | All providers | A | Covered | — | — |
| PROP-06 | **NCRYPT_KEY_USAGE_PROPERTY** | Signing / decrypt / key agreement flags | All providers | A | Covered | — | — |
| PROP-07 | **NCRYPT_ALGORITHM_GROUP_PROPERTY** | RSA / ECDSA / ECDH / EDDSA / AES / HMAC | All providers | A | Covered | — | — |
| PROP-08 | **NCRYPT_CHAINING_MODE_PROPERTY** | Read and write on symmetric keys | MS Software KSP (BCrypt) | A | Covered | — | — |
| PROP-09 | **NCRYPT_INITIALIZATION_VECTOR** | Read and write on symmetric keys | MS Software KSP (BCrypt) | A | Covered | — | — |
| PROP-10 | **NCRYPT_BLOCK_LENGTH_PROPERTY** | AES block size | MS Software KSP | B | Covered | — | — |
| PROP-11 | **NCRYPT_SECURITY_DESCR_PROPERTY** | Per-key ACL | MS Software KSP | B | Not covered | Returns NTE_NOT_SUPPORTED. PKCS#11 has no ACL model; would need a side-channel store mapping key labels to SDDL strings. | L |
| PROP-12 | **NCRYPT_UI_POLICY_PROPERTY** | Strong-key protection prompt | MS Software KSP; smart cards | B | Not covered | Depends on PromptUser (IFACE-07). Same UI-thread requirement. | L |
| PROP-13 | **NCRYPT_PIN_PROPERTY** | Per-key PIN | Smart card KSPs | B | Not covered | PIN is process-wide from SOFTHSM2_PIN. Per-key PINs would need a per-handle credential cache and re-login per operation. | M |

| ID | Feature | Detail | Market support | Ev | Status | Gap — what closing it would take | Eff |
|---|---|---|---|---|---|---|---|
| PROP-14 | **NCRYPT_CERTIFICATE_PROPERTY** | Certificate stored alongside the key | Smart card KSPs | B | Not covered | Would store a CKO_CERTIFICATE object with the same CKA_LABEL. SoftHSM2 supports certificate objects so this is straightforward. | M |
| PROP-15 | **NCRYPT_WINDOW_HANDLE_PROPERTY** | Parent HWND for dialogs | Smart card KSPs | B | Not covered | Only meaningful with PromptUser (IFACE-07). | L |
| PROP-16 | **NCRYPT_USE_COUNT_PROPERTY** | Key usage counter | Some HSMs | C | Not covered | Would need a counter object on the token incremented per operation. SoftHSM2 has no atomic counter primitive. | L |
| **KEY LIFECYCLE** | | | | | | | |
| LIFE-01 | **CreatePersistedKey** | Immediate and deferred generation | All providers | A | Covered | — | — |
| LIFE-02 | **FinalizeKey** | Deferred generation trigger | All providers | A | Covered | — | — |
| LIFE-03 | **OpenKey by name** | Asymmetric and symmetric | All providers | A | Covered | — | — |
| LIFE-04 | **DeleteKey** | Destroys all token objects | All providers | A | Covered | — | — |
| LIFE-05 | **EnumKeys** | Paged enumeration with state | All providers | A | Covered | — | — |
| LIFE-06 | **EnumAlgorithms** | NCryptEnumAlgorithms support | All providers | B | Not covered | Not in the KSP function table; ncrypt.dll answers from the registry. Verify our registration lists every algorithm we support or callers will not discover them. | S |
| LIFE-07 | **IsAlgSupported** | NCryptIsAlgSupported | All providers | B | Not covered | Same as LIFE-06: served from registry registration rather than a KSP entry point. Worth auditing register_ksp.ps1 against config.h. | S |
| LIFE-08 | **Machine vs user key scope** | NCRYPT_■MACHINE_■KEY_■FLAG | All providers | B | Not covered | All keys are token-global. PKCS#11 has no per-user scoping within a slot; would need label prefixing plus an ACL model. | M |
| **OPERATIONAL** | | | | | | | |
| OPS-01 | **Hardware key protection** | Tamper-resistant storage | All commercial providers | A | Not covered | Fundamental: SoftHSM2 is a software token using encrypted SQLite. Only fixable by pointing the KSP at a real HSM which the PKCS#11 abstraction already allows. | N/A |
| OPS-02 | **FIPS 140-2/3 validation** | Certified cryptographic module | All commercial providers | B | Not covered | Follows from OPS-01. Requires certified hardware and a formal validation programme. | N/A |
| OPS-03 | **Authenticode-signed binary** | Commercial code-signing certificate | All commercial providers | A | Not covered | Buy an OV or EV certificate and sign with signtool. No Microsoft involvement needed. See doc 10. | S |
| OPS-04 | **Multiple slots / tokens** | Selecting among several tokens | All commercial providers | B | Not covered | Only the first token-present slot is used. Add a SOFTHSM2_SLOT environment variable or a provider property and thread the slot ID through P11_CONTEXT. | M |
| OPS-05 | **Key attestation** | TPM / HSM-signed key origin proof | MS Platform Crypto; some HSMs | B | Not covered | No PKCS#11 standard mechanism exists. Vendor-specific and unreachable through a portable PKCS#11 layer. | L |
| OPS-06 | **Load balancing / HA** | Multiple HSM members | Thales; Entrust | C | Not covered | Handled inside vendor PKCS#11 libraries. Our KSP would inherit it automatically when pointed at such a library; nothing to implement. | N/A |
| OPS-07 | **Session pool concurrency** | Concurrent NCrypt calls | All providers | A | Covered | — | — |
| OPS-08 | **Re-initialisation without restart** | Recover from a bad DLL path | Vendor-specific | C | Not covered | InitOnceExecuteOnce is one-shot per process even on failure. Would need a resettable init guard; a real fix requires care around in-flight sessions. | M |
| OPS-09 | **PIN caching / re-login** | Session recovery after token logout | All commercial providers | C | Partial | C_Login tolerates CKR_■USER_■ALREADY_■LOGGED_■IN but there is no re-login path if a session drops. Add a retry that reopens and re-logs in on CKR_■USER_■NOT_■LOGGED_■IN. | M |
| OPS-10 | **Logging / diagnostics** | Trace output for support | All providers | A | Covered | — | — |

## Maintaining this document

The source of truth is `softhsm_ksp/docs/feature-matrix.csv`. One row per capability. To record that a gap has been closed, change that row's *status* to Covered and clear its *gap_solution*; to add a newly observed market feature, append a row with a fresh ID. Then regenerate:

```
python3 generate_feature_matrix.py
```

Every number on page 1 is computed from the CSV at generation time, so the summary cannot fall out of step with the table. Because the CSV is plain text, a change to coverage shows up as a one-line diff in review rather than an opaque binary change.

Actionable gaps by effort: **10 small**, **16 medium**, **18 large**. Rows marked N/A are excluded — they are deliberate positions (private key export is refused by design) or external blockers (CNG defines no EdDSA identifier; SoftHSM2 is a software token).

**A caution on the market columns.** An HSM datasheet is not a statement about its CNG provider. AWS CloudHSM performs AES, DES3 and HMAC in hardware, yet its KSP answers NCryptIsAlgSupported for exactly four identifiers. Every grade-C cell here carries that uncertainty. See docs/11-market-comparison.md for the full audit and its sources.